ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PRACTICE EXAM 8 (PAPER 2)

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET second paper (practical, 80 marks): design the circuit and flowchart of a smart dustbin, write its Arduino program, then add features to a Bluetooth robot-car program.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing (or another circuit program) for circuits, draw.io (or similar) for flowcharts and the Arduino IDE for programs. If you do not have a computer, draw the circuits and flowcharts neatly by hand and write the programs in your answer book.
3. Save your work regularly. Name every file with the question number and your name or examination number.
4. No circuits need to be built physically; designs are done on the computer or on paper.
5. Programs are not marked strictly on syntax: marks are given for correct logic, correct use of pins and functions, and comments.
6. Arduino Uno pins: digital pins 0–13 (PWM on 3, 5, 6, 9, 10 and 11), analogue inputs A0–A5, 5 V, 3,3 V and GND.

QUESTION 1

> A clinic wants a "no-touch" dustbin. When a hand is held in front of the bin, an IR obstacle sensor detects it and a servo motor opens the lid for 3 seconds. An ultrasonic sensor under the lid measures how full the bin is. When the bin is full, a red LED comes on, a buzzer beeps and the lid no longer opens.

The following components are available:

- Arduino Uno
- IR obstacle sensor module (VCC, GND, OUT – OUT goes LOW when an object is detected)
- HC-SR04 ultrasonic sensor (VCC, Trig, Echo, GND)
- SG90 servo motor
- Red LED and a 220 Ω resistor
- Active buzzer
- Breadboard and jumper wires
- 5 V power supply or USB cable

1.1 Using the given components, design the breadboard circuit for the smart dustbin. HINT: The circuit must include the correct wiring of the IR sensor output to a digital input; the HC-SR04 Trig and Echo pins to two digital pins; the servo signal to a PWM pin; the LED with its resistor and the buzzer on digital outputs; and correct 5 V and GND connections for all modules. (14)

1.2 Draw a flowchart that shows the logic of the smart dustbin. HINT: The flowchart must include setting up the pins and closing the lid; a loop that measures the fill level; a decision for "bin full" (closer than 8 cm) that switches on the LED and buzzer and keeps the lid closed; a decision for "hand detected" that opens the lid for 3 seconds and closes it; and a check for an impossible distance reading. (14)

1.3 Save and print (or neatly present) the breadboard circuit and the flowchart. (2)

[30]

QUESTION 2

Write the program for the smart dustbin in QUESTION 1. Use these connections: IR sensor OUT to pin 2, HC-SR04 Trig to pin 7 and Echo to pin 8, servo signal to pin 9, red LED to pin 12 and buzzer to pin 13. The bin is 40 cm deep.

2.1 Write an Arduino program that does the following: (18)

• Sets up all the pins and the servo, starts the serial monitor and closes the lid (0°) at the start.

• Measures the distance from the ultrasonic sensor to the rubbish every time the loop runs.

• Prints the fill level as a percentage, using: fill % = (40 − distance) ÷ 40 × 100.

• When the distance is less than 8 cm, switches on the red LED, beeps the buzzer three times and does not open the lid.

• Otherwise switches the LED off and, when a hand is detected, opens the lid (90°) for 3 seconds and then closes it.

• Prints "Sensor error" if the distance is 0 or more than 100 cm, and keeps the lid closed.

• Is well commented, so the logic is clear.

2.2 Save and print (or neatly present) the program. (2)

[20]

QUESTION 3

The program below drives a two-motor robot car with an L298N motor driver. A cellphone app sends single letters through an HC-05 Bluetooth module, which is connected to the Arduino's serial pins: F = forward, B = back, L = left, R = right, S = stop.

```
// Bluetooth robot car
const int enA = 5, in1 = 6, in2 = 7;     // left motor
const int enB = 10, in3 = 8, in4 = 9;    // right motor
int speed = 200;

void setup() {
  Serial.begin(9600);                    // HC-05 on the serial pins
  pinMode(enA, OUTPUT); pinMode(in1, OUTPUT); pinMode(in2, OUTPUT);
  pinMode(enB, OUTPUT); pinMode(in3, OUTPUT); pinMode(in4, OUTPUT);
}

void drive(int l1, int l2, int r1, int r2) {
  digitalWrite(in1, l1); digitalWrite(in2, l2);
  digitalWrite(in3, r1); digitalWrite(in4, r2);
  analogWrite(enA, speed);
  analogWrite(enB, speed);
}
```

```
void loop() {
  if (Serial.available() > 0) {
    char c = Serial.read();
    if (c == 'F') drive(HIGH, LOW, HIGH, LOW);
    if (c == 'B') drive(LOW, HIGH, LOW, HIGH);
    if (c == 'L') drive(LOW, HIGH, HIGH, LOW);
    if (c == 'R') drive(HIGH, LOW, LOW, HIGH);
    if (c == 'S') drive(LOW, LOW, LOW, LOW);
  }
}
```

3.1 Add three speed commands: '1' sets the speed to 100, '2' sets it to 170 and '3' sets it to 255. The new speed must be used from the next movement command onwards. (9)

3.2 For safety, make the car stop automatically if no command has been received for 2 seconds (for example, when the phone goes out of range). Use millis(). (9)

3.3 Add a horn: a buzzer on pin 4 must sound for half a second when the letter 'H' is received. (9)

3.4 Save and print (or neatly present) your final program. (3)

[30]

TOTAL: 80

END OF QUESTION PAPER